

A Level Mathematics 9709 · M2 Mechanics Extension (legacy)

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 7 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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2. 变加速度 | Variable acceleration
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5. 平衡与力矩 | Moments and equilibrium
6. 弹性弦与胡克定律 | Elastic strings
7. M2拓展综合 | Extension synthesis

第 1 课 | M2定位与M1复习 | Positioning and prerequisites

Lesson 1 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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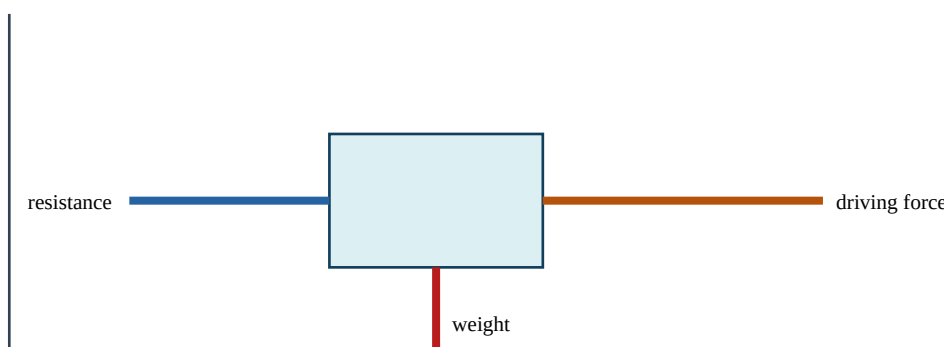
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

先熟练向量、微积分、 $F=ma$ 与能量，再学习更复杂模型。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

速度 $v=3t^4$ ，求 a 与 $t=2$ 时位移增量框架。

$a=dv/dt=6t^3$ ；位移需积分 v 并用区间端点。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | M2定位与M1复习 | Positioning and prerequisites

例题二：Cambridge 风格应用 | Cambridge-style application

Why must a model state assumptions?

Assumptions define which physical effects are included and when formulas are valid.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

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Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：速度 $v=3t+4t$ ，求 a 与 $t=2$ 时位移增量框架。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $a=dv/dt=6t+4$ ；位移需积分 v 并用区间端点。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why must a model state assumptions?

独立完成后对照：Assumptions define which physical effects are included and when formulas are valid.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 变加速度 | Variable acceleration

Lesson 2 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

一维运动满足 $v=ds/dt$ 、 $a=dv/dt=v\,dv/ds$ 。

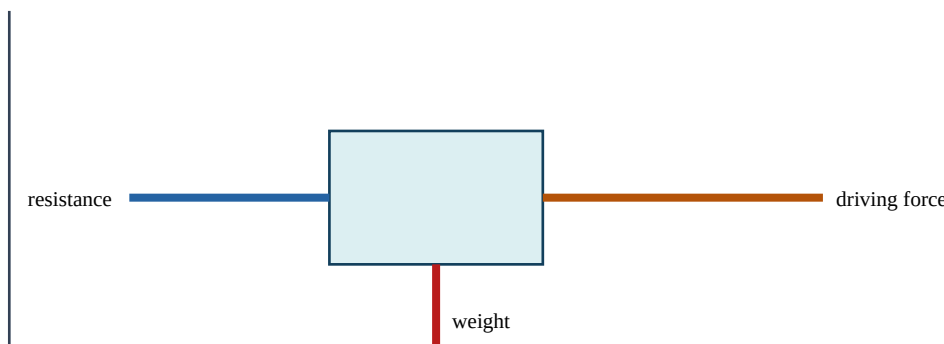
English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

根据已知变量选择对 t 或 s 积分，并使用初始条件。

把知识点拆成可执行步骤 | A reliable method

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$dv/dt=6t$, $v=3t+C$; $C=2$ 。

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第 2 课（续） | 变加速度 | Variable acceleration

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Given $a=2s$ and $v=3$ at $s=0$, find v .

$v \, dv/ds=2s$; integrate: $v^2/2=s+C$, $C=9/2$, hence $v=2s+9$.

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Given $a=2s$ and $v=3$ at $s=0$, find v .

独立完成后对照： $v\,dv/ds=2s$; integrate: $v^2/2=s+C$, $C=9/2$, hence $v=2s+9$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 圆周运动 | Circular motion

Lesson 3 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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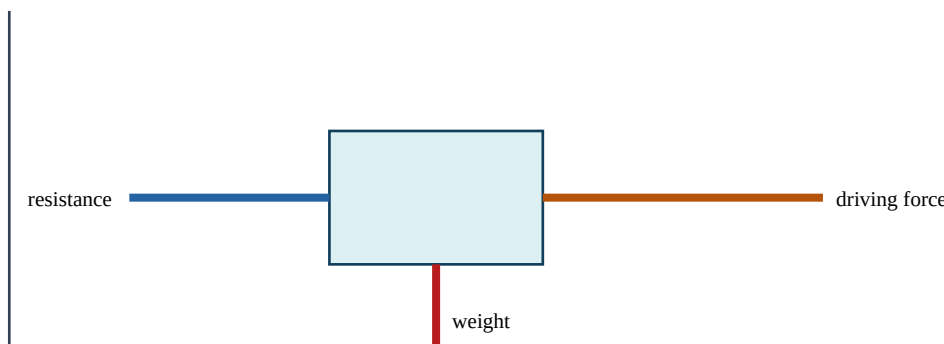
English core explanation. A diagram is part of the reasoning, not decoration. Label known values, identify the theorem and its conditions, then connect each algebraic line to the geometry. Length, area and volume scale by k , k and k respectively.

为什么方法成立 | Reasoning behind the method

画径向受力图并以向心方向为正，列 $\Sigma F = mv^2/r$ 。

把知识点拆成可执行步骤 | A reliable method

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例题一：从题意到完整解答 | Worked example 1

0.5 kg 物体速度6、半径2，求径向合力。

$$F = 0.5 \times 36/2 = 9 \text{ N}.$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续） | 圆周运动 | Circular motion

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What provides centripetal force for a car on a flat bend?

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第三个迁移任务 | Transfer task

What provides centripetal force for a car on a flat bend?

独立完成后对照：Horizontal friction between tyres and road.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 质心 | Centre of mass

Lesson 4 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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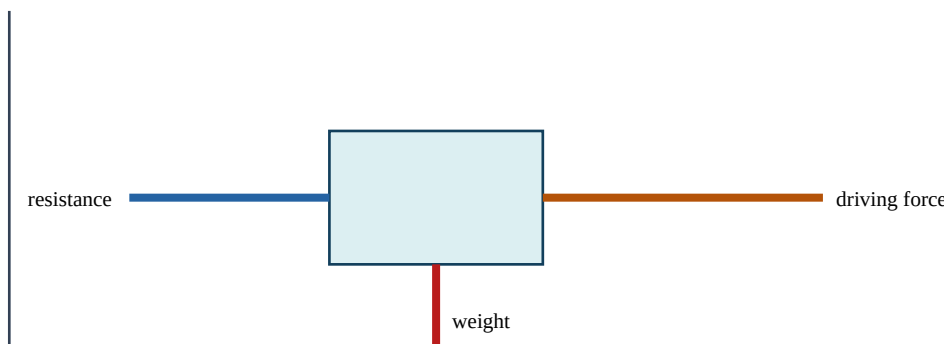
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为什么方法成立 | Reasoning behind the method

选原点与坐标轴，列 $x = \Sigma mx / \Sigma m$ 。

把知识点拆成可执行步骤 | A reliable method

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5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

2 kg在 $x=1$, 3 kg在 $x=5$, 求质心。

$$x = (2 \times 1 + 3 \times 5) / 5 = 3.4。$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 质心 | Centre of mass

例题二：Cambridge 风格应用 | Cambridge-style application

Why can symmetry locate a centre of mass?

For uniform density, equal mass is distributed symmetrically on both sides of each symmetry axis.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

质心位置是质量加权平均；均匀薄片可用对称与分割。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。选原点与坐标轴，列 $x = \Sigma mx / \Sigma m$ 。

Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：2 kg在 $x=1$ ，3 kg在 $x=5$ ，求质心。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $x = (2 \times 1 + 3 \times 5) / 5 = 3.4$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why can symmetry locate a centre of mass?

独立完成后对照：For uniform density, equal mass is distributed symmetrically on both sides of each symmetry axis.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 平衡与力矩 | Moments and equilibrium

Lesson 5 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

刚体平衡需要合力为0且任一点合力矩为0。

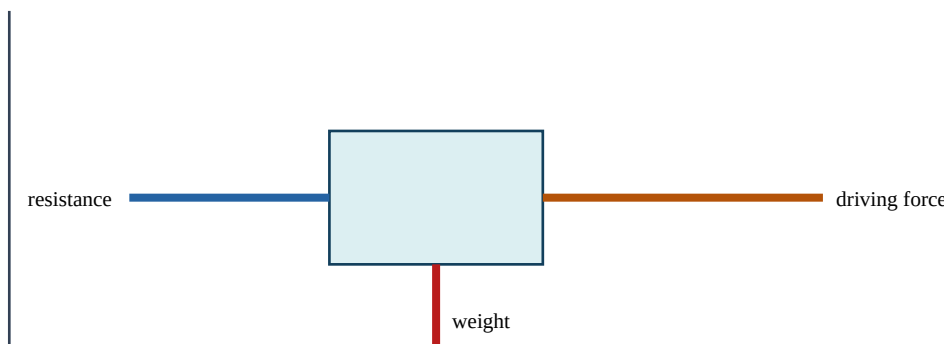
English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

力矩=力 $\times$ 垂直距离；统一顺时针符号。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

4 m均匀梁重200 N，中点支撑，右端挂50 N，左端需挂多大力平衡？

关于中点： $F \times 2 = 50 \times 2$ ，所以 $F = 50$ N。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 平衡与力矩 | Moments and equilibrium

例题二：Cambridge 风格应用 | Cambridge-style application

State two equilibrium conditions.

Resultant force is zero and resultant moment about any point is zero.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

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把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

刚体平衡需要合力为0且任一点合力矩为0。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。力矩=力×垂直距离；统一顺时针符号。

Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：4 m均匀梁重200 N，中点支撑，右端挂50 N，左端需挂多大力平衡？

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：关于中点： $F \times 2 = 50 \times 2$ ，所以 $F=50$ N。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

State two equilibrium conditions.

独立完成后对照：Resultant force is zero and resultant moment about any point is zero.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 弹性弦与胡克定律 | Elastic strings

Lesson 6 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

轻弹性弦张力 $T = \lambda x/l$, 弹性能 $= \lambda x/(2l)$, 仅在拉伸且弹性限度内。

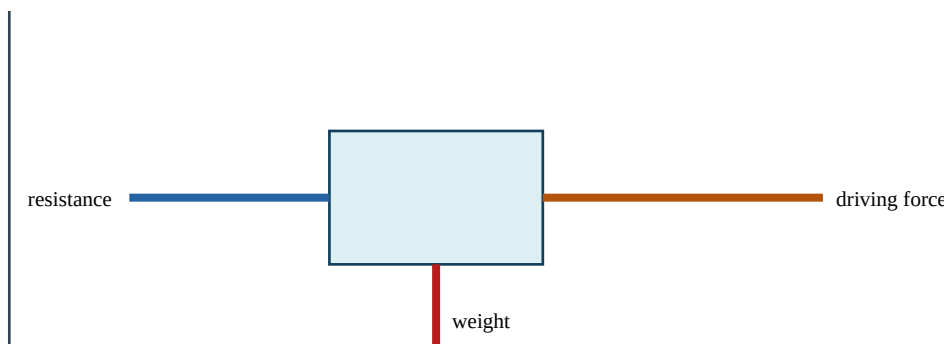
English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

先分清自然长 l 与伸长 x ; 松弛时张力为 0。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$\lambda = 60 \text{ N}$, 自然长 2 m , 伸长 0.3 m , 求张力。

$T = 60 \times 0.3/2 = 9 \text{ N}$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 弹性弦与胡克定律 | Elastic strings

例题二：Cambridge 风格应用 | Cambridge-style application

Find elastic potential energy.

$$E = 60 \times 0.3 / (2 \times 2) = 1.35 \text{ J.}$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

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Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, $F=ma$ and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $\lambda = 60 \text{ N}$ ，自然长 2 m ，伸长 0.3 m ，求张力。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $T = 60 \times 0.3/2 = 9 \text{ N}$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Find elastic potential energy.

独立完成后对照： $E = 60 \times 0.3/(2 \times 2) = 1.35 \text{ J}$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | M2拓展综合 | Extension synthesis

Lesson 7 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

复杂问题通常分阶段并组合能量、牛顿定律和几何约束。

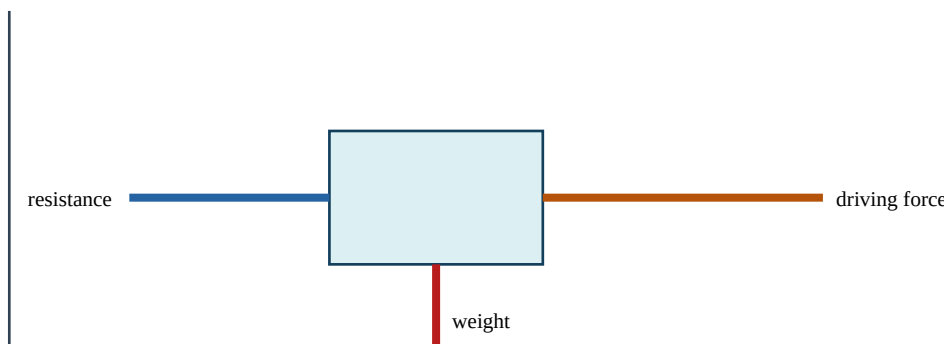
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

每个事件点更新速度、张力和接触状态；验证 $R \geq 0$ 与绳是否拉紧。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

法向反作用力算成负数表示什么？

原先“保持接触”假设已失效，物体会在更早位置失去接触。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | M2拓展综合 | Extension synthesis

例题二：Cambridge 风格应用 | Cambridge-style application

Why should legacy content be kept separate?

It extends mathematical mechanics but should not be mistaken for a current examination component.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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再推一步 | Extension and connection

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

复杂问题通常分阶段并组合能量、牛顿定律和几何约束。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。每个事件点更新速度、张力和接触状态；验证 $R \geq 0$ 与绳是否拉紧。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：法向反作用力算成负数表示什么？

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：原先“保持接触”假设已失效，物体会在更早位置失去接触。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why should legacy content be kept separate?

独立完成后对照：It extends mathematical mechanics but should not be mistaken for a current examination component.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?